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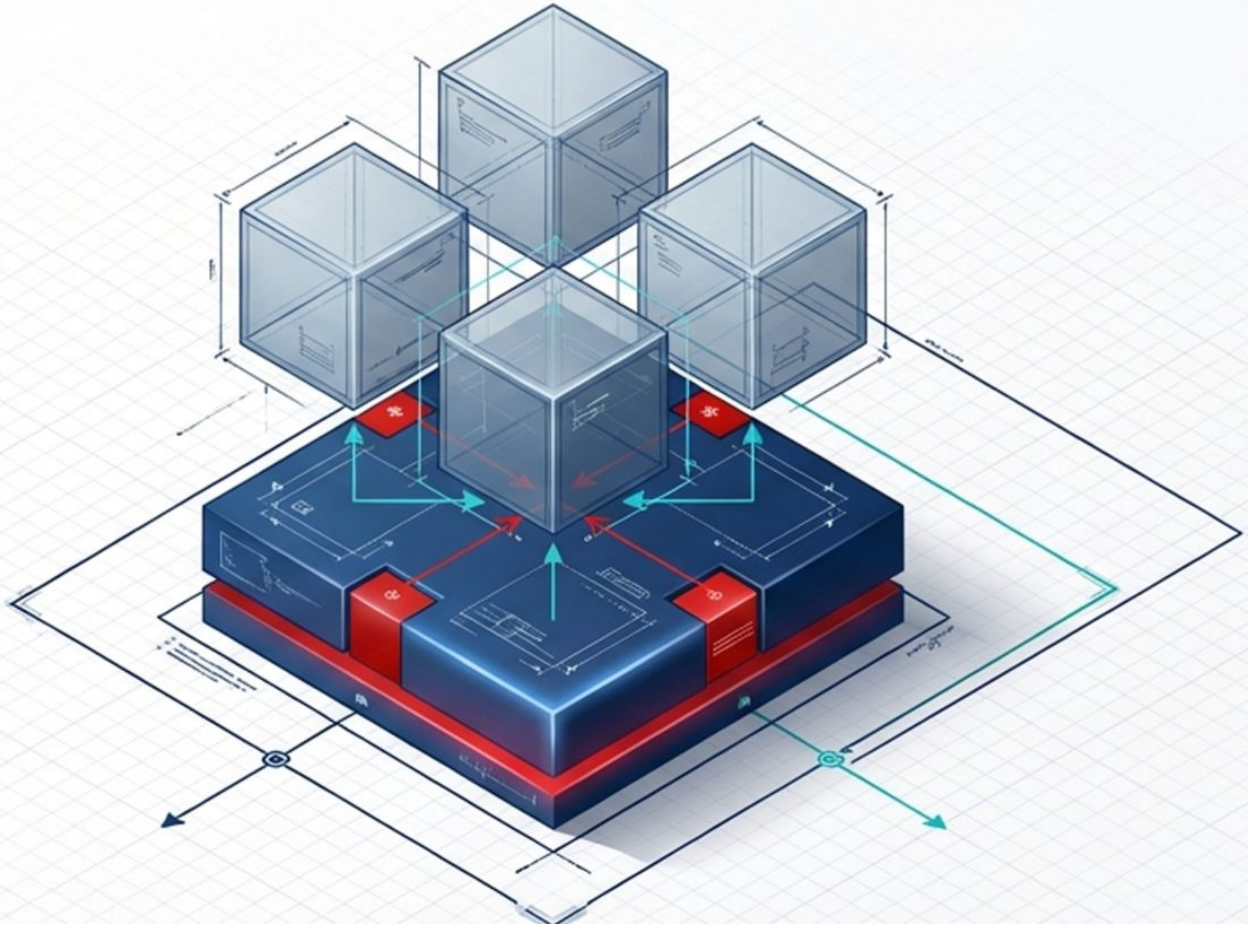
Persistent Storage in OpenShift

CSI, SDS, and Moving Beyond VMware vSAN

Red Hat

THE DATA DILEMMA IN A STATELESS WORLD

Demystifying OpenShift Persistent Storage Architecture



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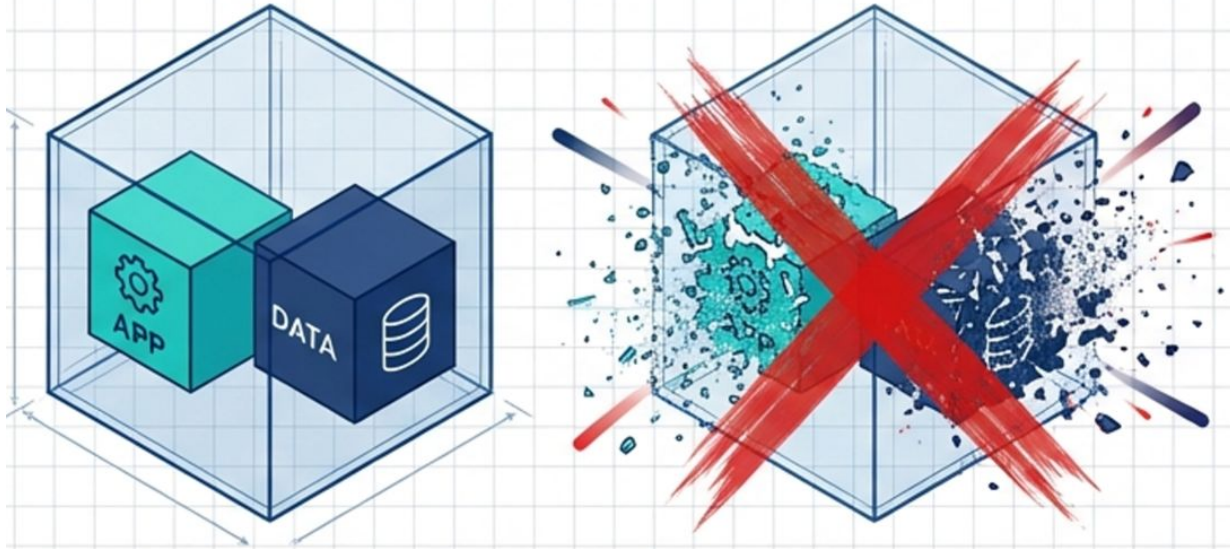
THE EPHEMERAL PROBLEM

Containers are ephemeral by design. When a pod is destroyed, any data written to its default file system is permanently deleted.

THE POD LIFECYCLE

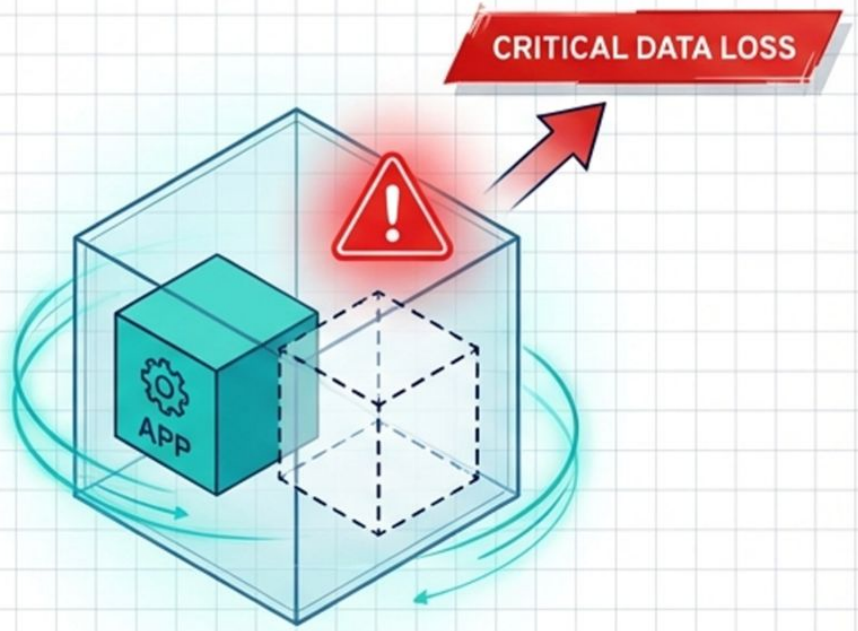
STEP 1: POD CREATION

STEP 2: POD TERMINATION



OPERATIONAL

THE CONSEQUENCE



NEW POD STARTS WITH APP RESTORED,
BUT DATA IS PERMANENTLY GONE.

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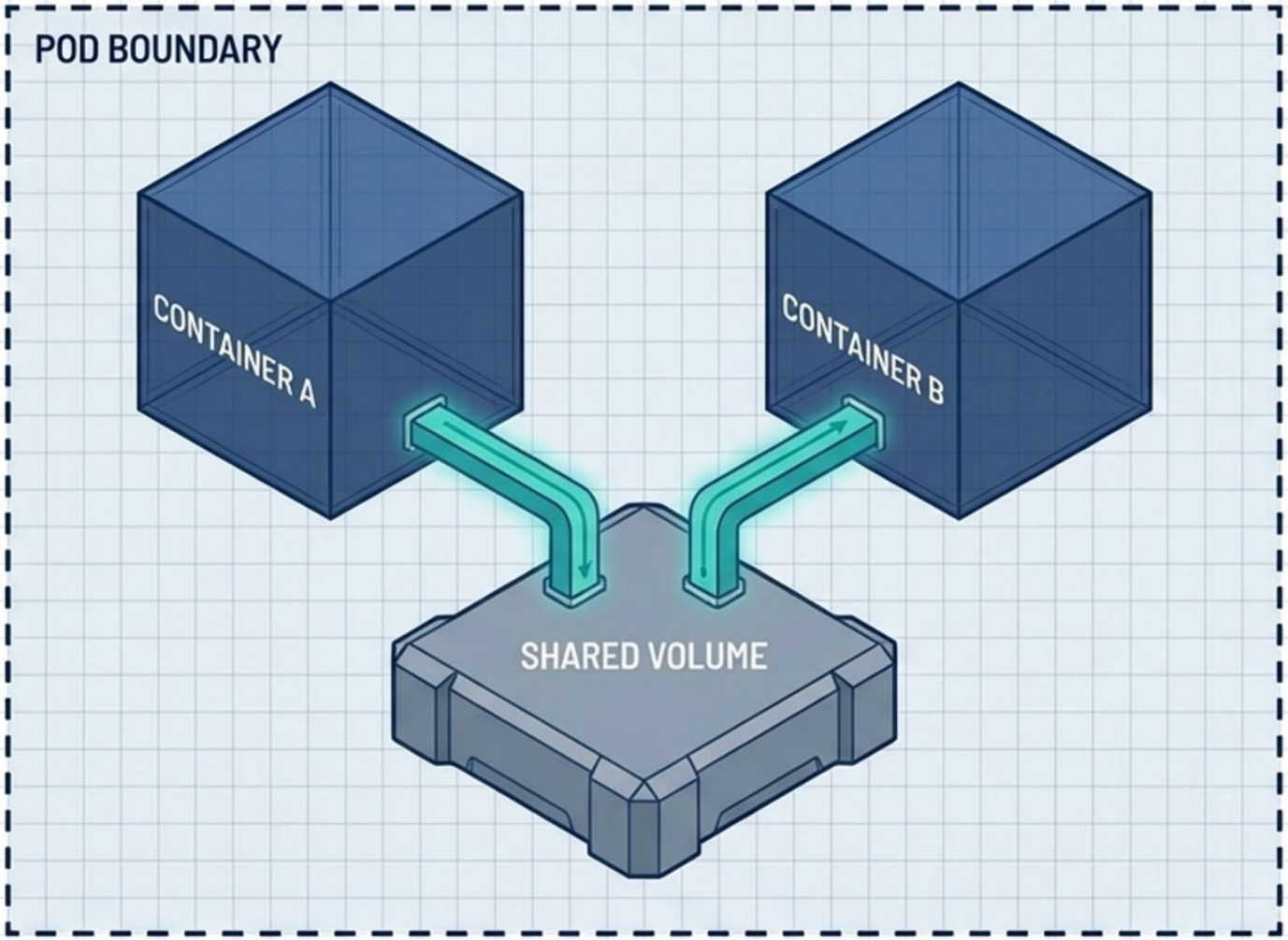
THE INITIAL SOLUTION: REGULAR VOLUMES

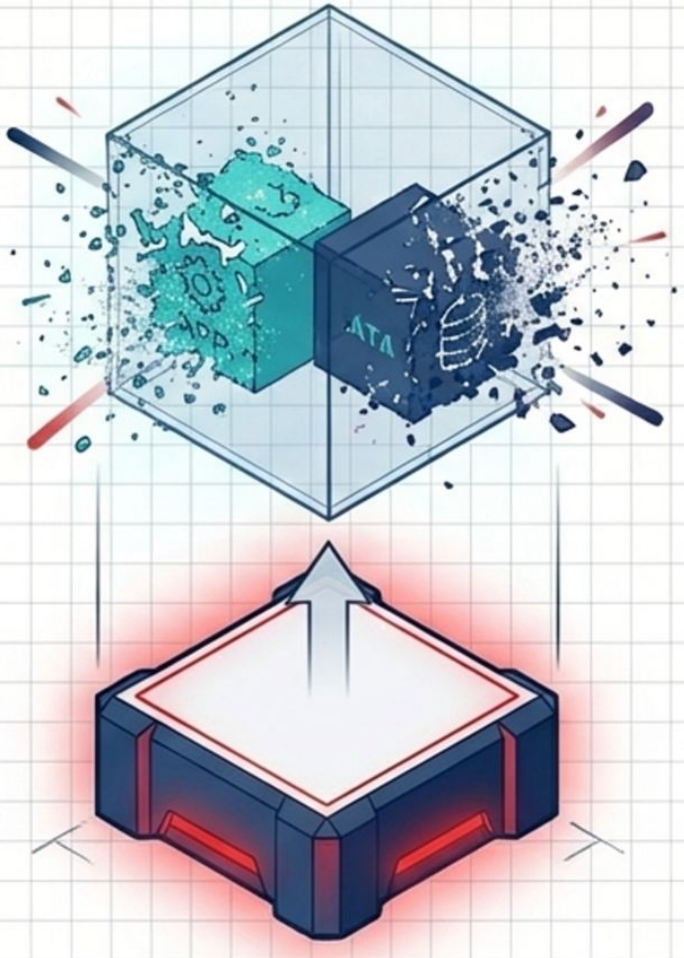


KEY TAKEAWAY: REGULAR VOLUMES OFFER SHARED STORAGE BETWEEN CONTAINERS IN A POD, BUT THEY STILL DIE WITH THE POD.

- Solves container-level isolation.
- Allows reliable mounting of the same file system for multiple services in one pod.
- The Catch: The volume's lifecycle remains strictly bound to the pod's lifecycle.

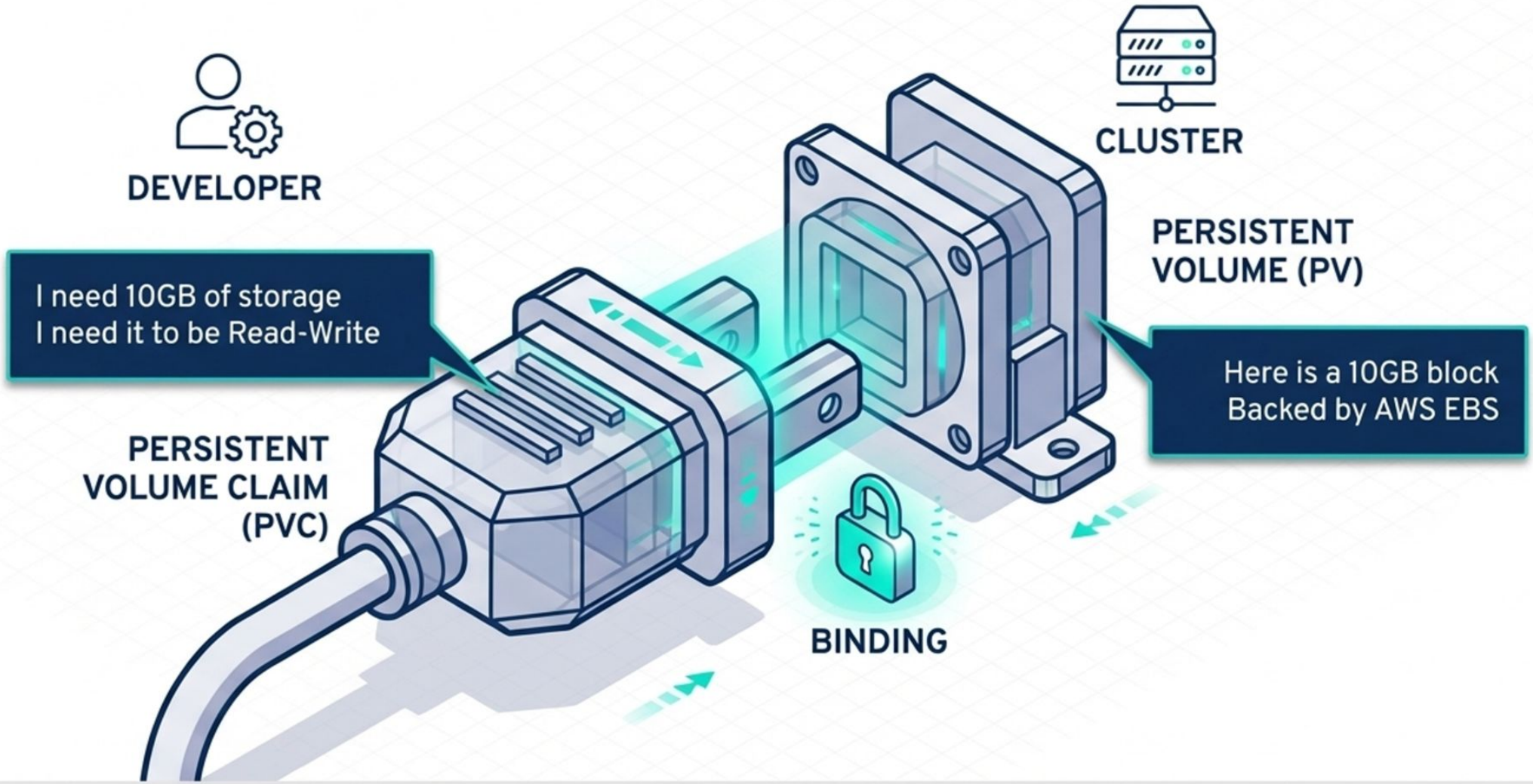
BREAKING THE LIFECYCLE TIE: PERSISTENT VOLUMES

FEATURE	REGULAR VOLUME	PERSISTENT VOLUME (PV)
Scope	Pod-level	Cluster-level
Lifecycle	Dies with the Pod	Outlives the Pod (Independent)
Target Workload	Scratch space, shared caches	Databases, VM disks, stateful apps
Management Persona	Developer	Cluster Administrator



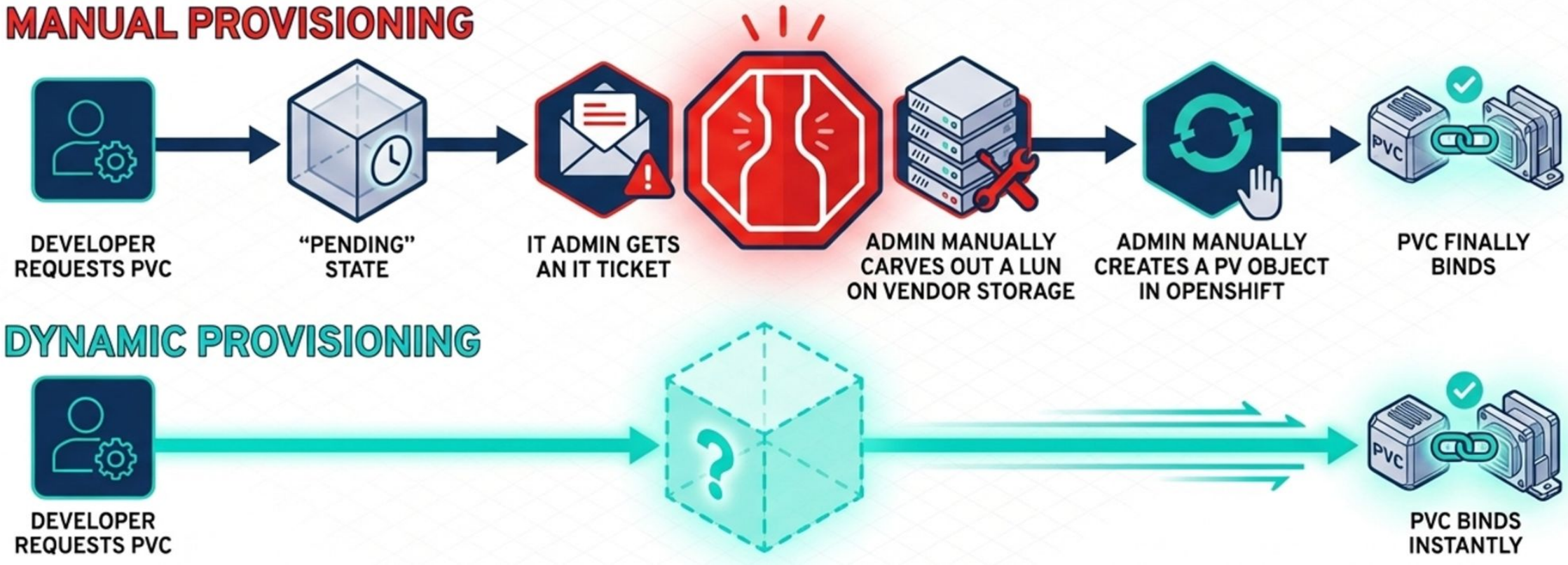
PV (PERSISTENT VOLUME)

THE HANDSHAKE: PERSISTENT VOLUME CLAIMS (PVCS)



THE BOTTLENECK: MANUAL PROVISIONING

How do we eliminate the IT ticket bottleneck and provision storage on-demand?



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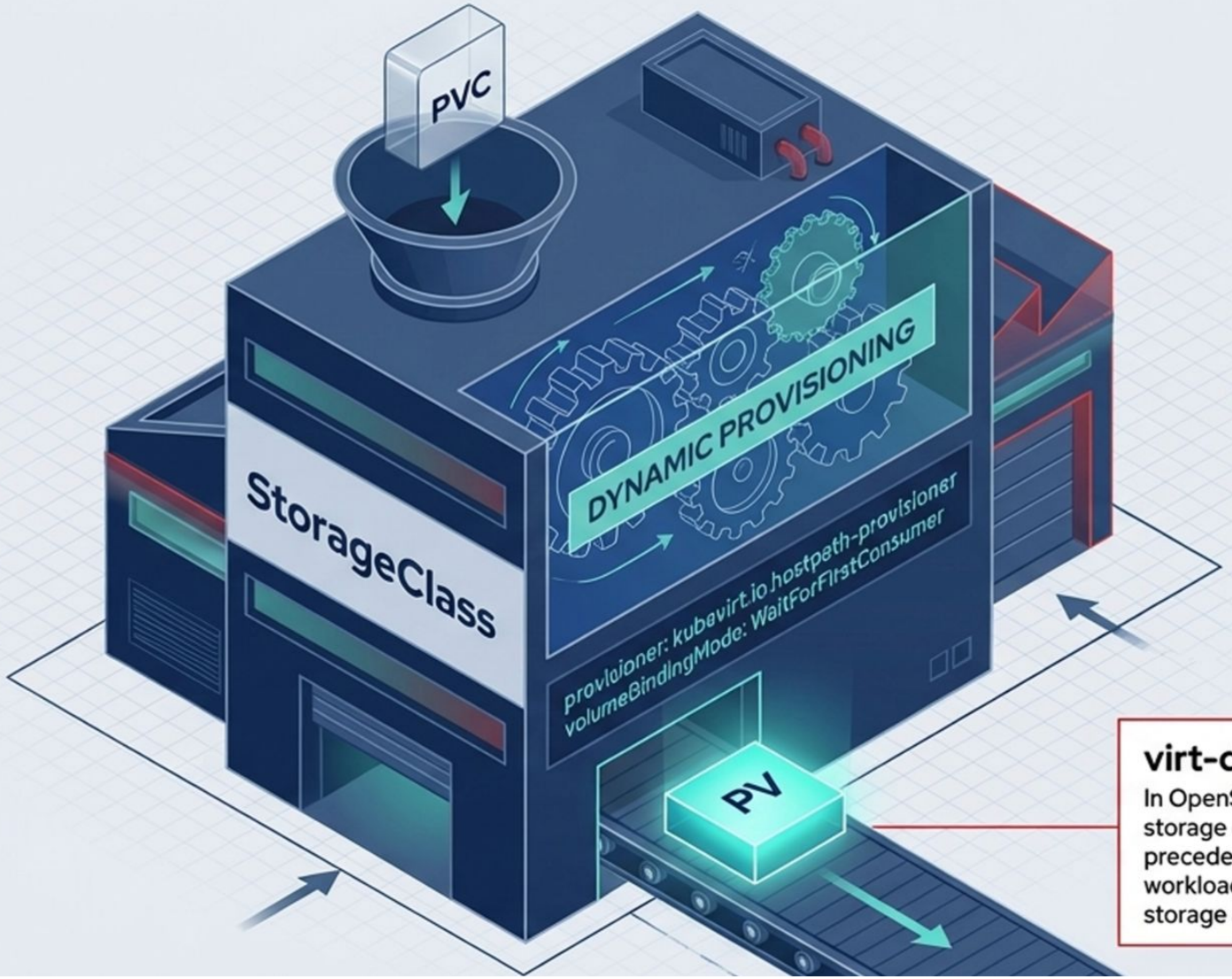
THE AUTOMATION ENGINE: STORAGE CLASSES

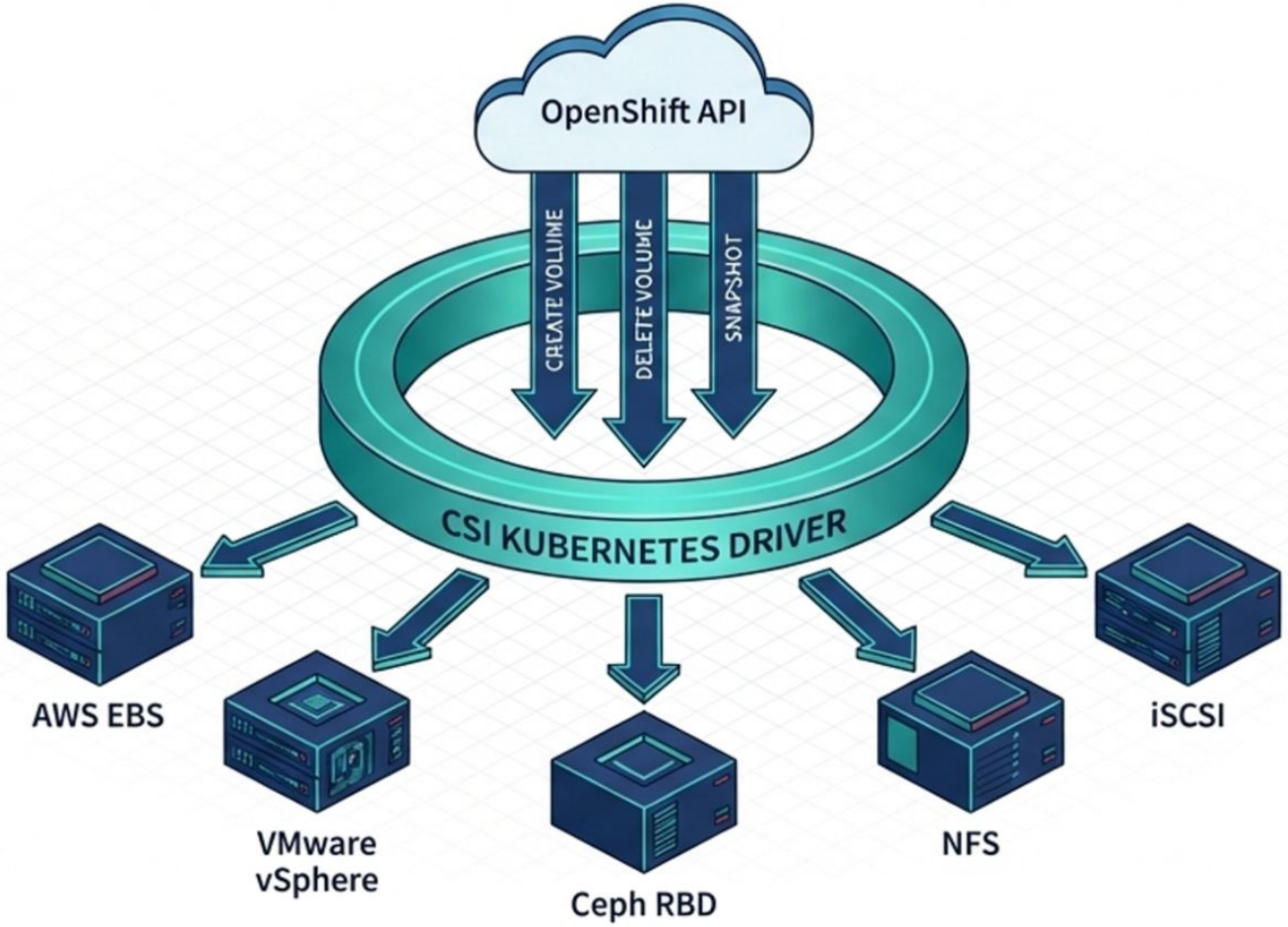


virt-default

In OpenShift Virtualization, a specific storage class can be annotated to take precedence specifically for virtualization workloads, ensuring VMs get optimized storage automatically.

THE UNIVERSAL TRANSLATOR: CONTAINER STORAGE INTERFACE (CSI)

The CSI decouples core OpenShift from specific vendor implementations, creating a standard industry interface.



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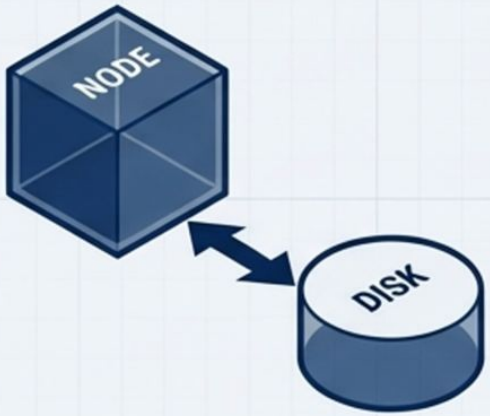
FINE-TUNING THE CONNECTION: ACCESS MODES

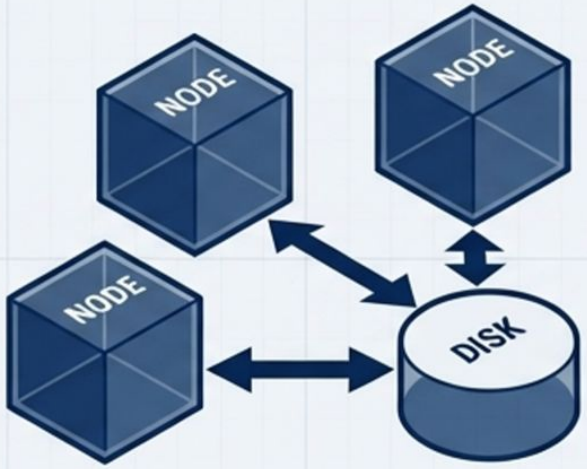
ReadWriteOnce (RWO)



Definition: Mounted as read-write by a single node.

Typical Use Case: Standard databases, single-instance apps.

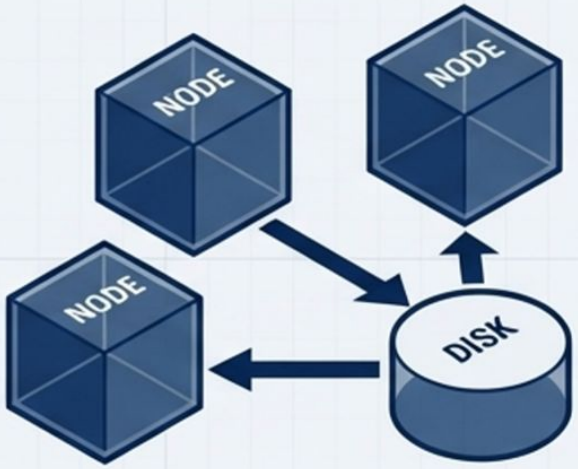
ReadWriteMany (RWX)



Definition: Mounted as read-write by many nodes simultaneously.

Typical Use Case: Shared web roots, OpenShift Virtualization Live Migration.

ReadOnlyMany (ROX)



Definition: Mounted as read-only by many nodes.

Typical Use Case: Configuration data, shared static assets.

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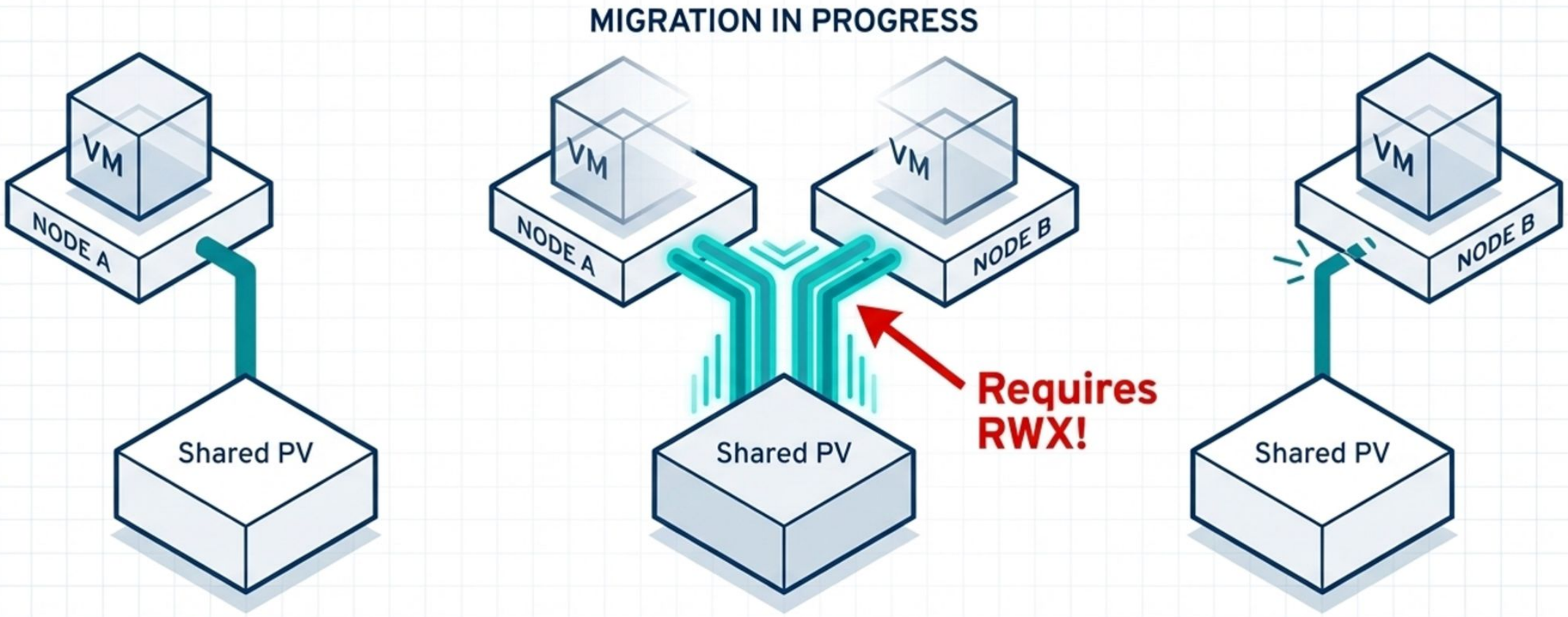
THE LIVE MIGRATION CHALLENGE



If the VMs that are sharing disks are running on different nodes, ReadWriteMany (RWX) is an absolute requirement for Live Migration to succeed.

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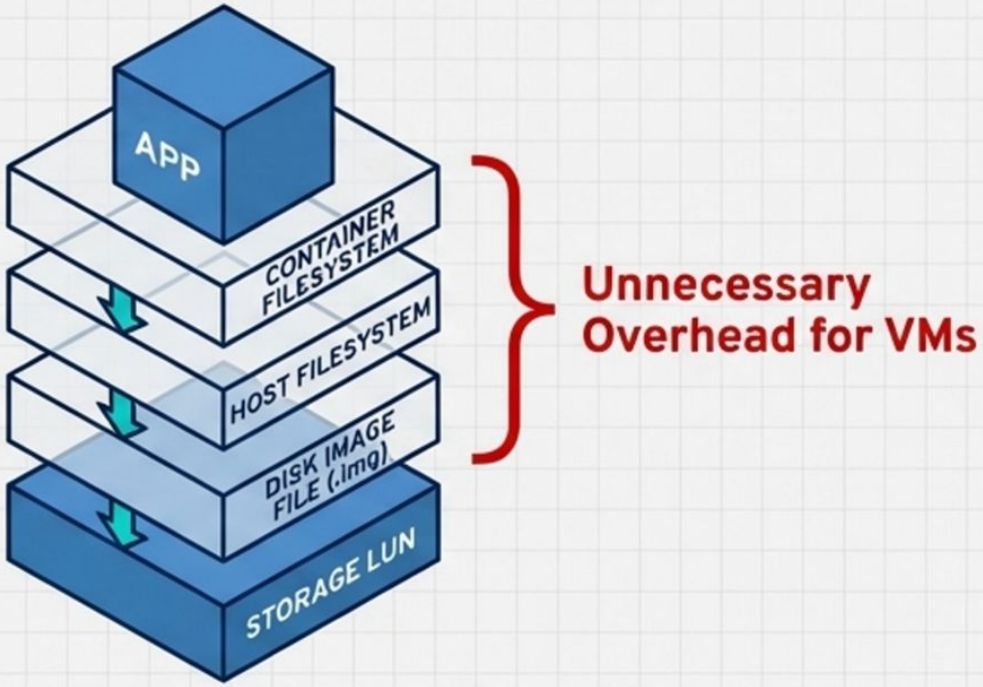
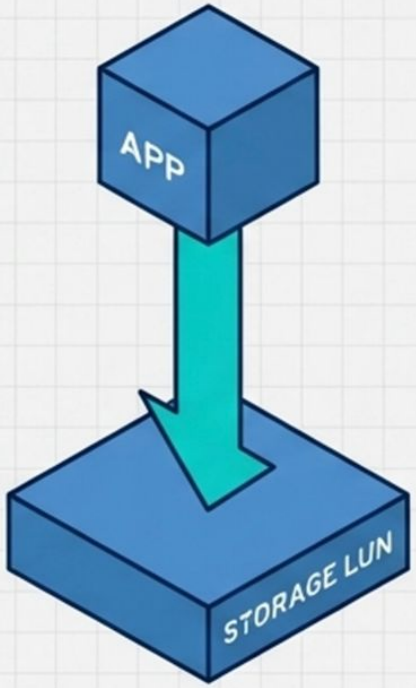
ELIMINATING OVERHEAD: VOLUME MODES

FILESYSTEM MODE



BLOCK MODE



Filesystem:	Stores virtual disk on a file system-based volume. Good for generic containers.
Block:	Stores virtual disk directly on the block volume. Crucial for VM performance.

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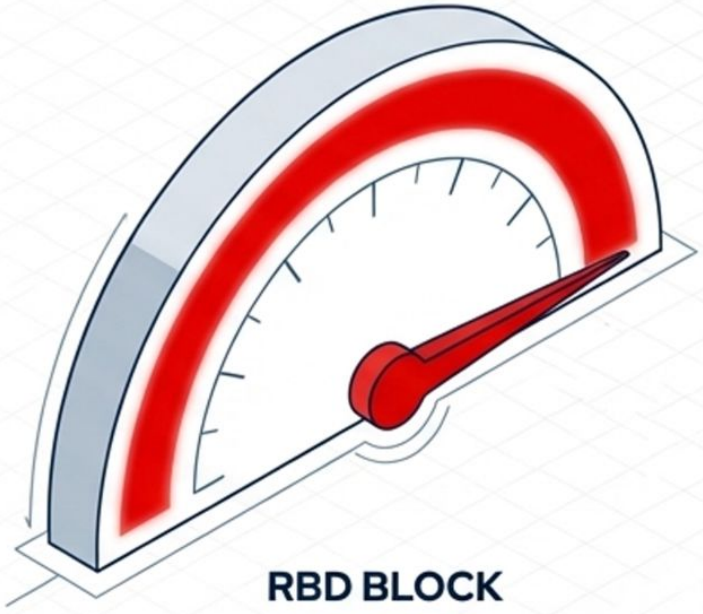
OPTIMIZED VIRTUALIZATION STORAGE WITH ODF

> When using OpenShift Virtualization with Red Hat OpenShift Data Foundation, specify RBD block mode persistent volume claims (PVCs)... RBD block mode volumes are more efficient and provide better performance.

CEPHFS / RBD FILESYSTEM PVCs



RBD BLOCK MODE PVCs

REQUIRED PARAMETERS:

- **Storage Class:** ocs-storagecluster-ceph-rbd
- **VolumeMode:** Block

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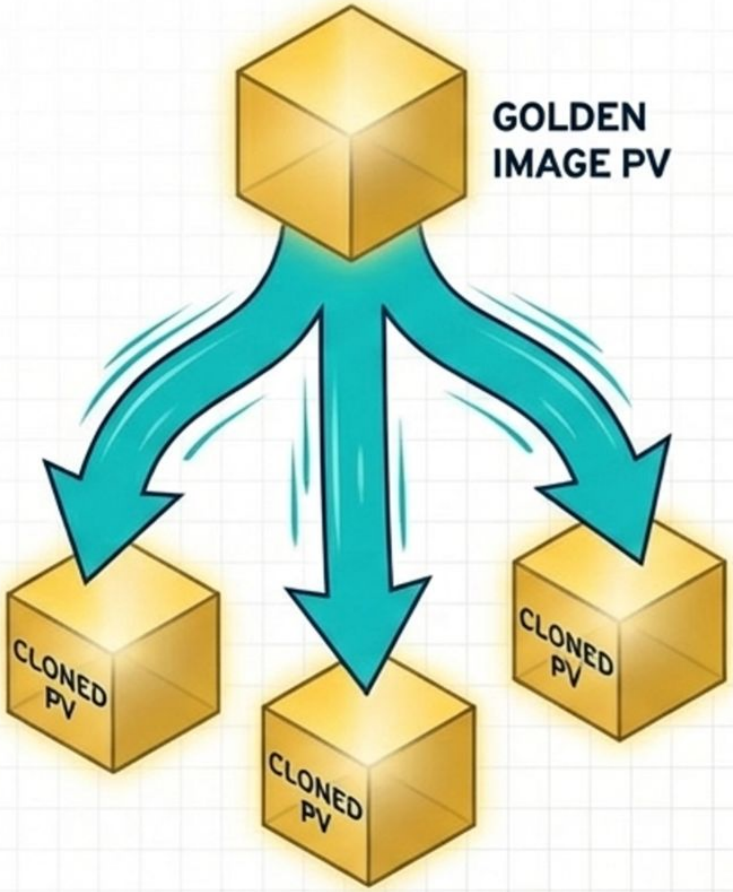
ADVANCED CAPABILITIES: CSI VOLUME CLONING



KEY REQUIREMENTS:

- CSI driver must support volume cloning.
- Source and target PVCs must share the same StorageClass.
- Source and target PVCs must share the same VolumeMode.

Smart-Cloning

Uses underlying storage snapshots (e.g., ODF).

Extremely fast, highly efficient.

Host-Assisted

Streams data node-to-node.

Slower, used as a fallback.

Note: Smart-cloning cannot clone across different volume modes.

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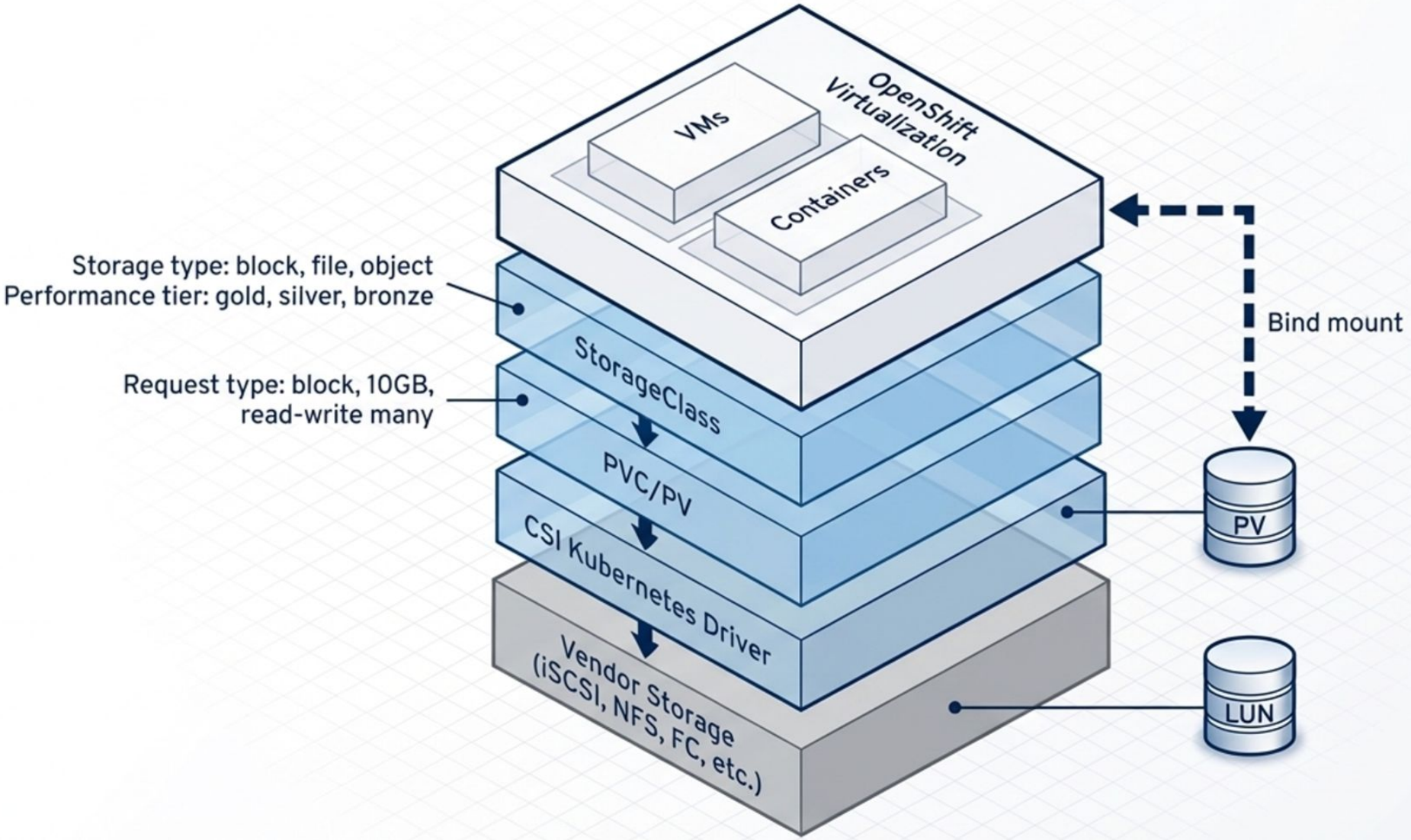
THE COMPLETE STORAGE BLUEPRINT



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ARCHITECTURAL IMPERATIVES

1. Decouple for Resilience

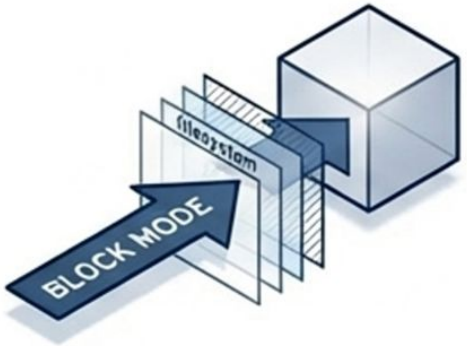
Always use PVs and PVCs to separate the lifecycle of your stateful data from the ephemeral nature of your pods.

2. Automate Everything

Rely on StorageClasses and dynamic provisioning to eliminate manual LUN management and IT IT bottlenecks.

3. Optimize for the Workload

Use RWX Access Modes for live migration support, and Block Volume Modes to eliminate filesystem overhead for Virtual Machines.



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Thank You

Questions? See OpenShift Storage and CSI documentation on docs.redhat.com.

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